

DIPTANSU PODDAR

3rd Year Undergraduate, Chemical Engineering
Indian Institute of Technology Kanpur

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My Github | diptansu-poddar

EDUCATION

Year	Degree	Institute	CPI/%
2021 - 2025	B.Tech, CHE	Indian Institute of Technology, Kanpur	9.04/10
2021	Class XI - XII (ISC)	Calcutta Boys' School, Kolkata	97.25%
2019	Class X (ICSE)	Calcutta Boys' School, Kolkata	97%

SCHOLASTIC ACHIEVEMENTS

- Secured **AIR 6050** in JEE Advanced 2021 among 0.2 million and **AIR 7829** in JEE Main 2021 among 1.2 million candidates
- Received **2 Awards for Academic Excellence** from IIT Kanpur in 2021 and 2022 for being among the **top 10 percentile**
- Received **A*** grade for **exceptional performance** in **ESC101 (Fundamentals of Computing)** and **CHE211 (Fluid Mechanics)**
- Successfully **changed branch** to Chemical Engineering from Materials Science and Engineering after 1st year
- Qualified for **INSPIRE** scholarship of DST, GoI by being among the top 1% in the ISC Examination (Class XII)

RESEARCH PROJECT

- CFD-DEM Simulation** | *Mentor: Anurag Tripathi, Associate Prof., Dept. of Chemical Engineering* Sept 2022 - Dec 2022
 - Simulation study of the flow of granular particles in a chute using C++
 - Incorporated fluid dynamics components such as packing fraction and viscosity in the simulation study

KEY PROJECTS

- A Mock Chemical Startup** | *Course Project, CHE 261: Chemical Process Industries* Jan 2022 - May 2022
 - Established a mock company to produce Diethyl phthalate in India after conducting detailed feasibility and cost analysis
 - Analyzed possible profit margins by enquiring about actual raw material/product prices and cost of required equipments
 - Researched on the steps and processes needed to manufacture the chemical from research papers and patents filed
 - Demonstrated with block diagrams how a chemical plant can produce 1000 kg/day of Diethyl phthalate after conducting appropriate material balances on input and output products, identifying unit operations and operating conditions
- Intro to Machine Learning** | *Chemineers Society* Dec 2021- March 2022
 - Built a Python neural network model using **TensorFlow and Keras** to predict relative humidity using 13 regressors
 - Performed **exploratory data analysis**, dropped redundant rows and dealt with categorical data such as dates and times
 - Refined the model by varying the number of nodes and layers and achieved a mean absolute error of only 2.62
- ClassifyMeister** | *Stamatics Society* May 2023 - June 2023
 - A learning project to study classification algorithms such as KNN, SVM, regression, Naive Bayes, and Random Forest
 - Designed a binary **credit score evaluation** model based on personal features and habits using a data-set with 160+ entries
 - Analyzed the use of Decision Tree and Random Forest algorithms in a research work on main transformer fault evaluation

TECHNICAL SKILLS

- Programming Skills:** C, C++, Python, Java
- Miscellaneous:** Git, Colab, Jupyter notebook, MS Excel
- Frameworks/Libraries:** NumPy, pandas, scikit-learn, matplotlib, TensorFlow

RELEVANT COURSES

#upcoming

Chem. Eng. Thermodynamics (A)	Fluid Mechanics and its Appl. (A*)	Thermodynamics (A)
Intro. to Chem. Eng. and Process Calculations	Chemical Process Industries	Computational Methods in Eng.
Fundamentals of Computing (A*)	Data Structure and Algorithm [#]	Linear Algebra and ODE
Introduction to Logic (A)	Electrodynamics (Physics II) (A)	Introduction to Electronics (A)

CERTIFICATION

- Training Program** | *Dept. of Electrical Engineering, IIT Kanpur (online)* May 2023
 - IIT Kanpur Intensive Training School (ITS) on PYTHON for Data Science, Machine Learning and Deep Learning

POSITIONS OF RESPONSIBILITY

- Manager (previously Secretary), Research Wing** | *Academics and Career Council, IIT Kanpur* May 2022 - Present
 - Recruited and led** a team of 14 secretaries to smoothly perform all activities of the research wing
 - Conducted information sessions** for students desirous of pursuing higher studies and internships at foreign universities
- Seceratry, Book Club** | *Media and Cultural Council, IIT Kanpur* May 2022 - April 2023
 - Wrote book recommendations, designed posters, organized book discussions as part of our club activity
- Academic Mentor, PHY103** | *Counselling Service, IIT Kanpur* May 2022 - April 2023
 - Mentored freshers seeking help and conducted academic classes for first year courses

EXTRACURRICULAR ACTIVITIES

- 5th Inter-IIT Cultural Meet** | *IIT Madras* Jan 23
 - Took part in the following events: Tag-team: a culinary challenge, STEW: a verbal battle of wit, and India Quiz
- Galaxy 2023** | *Inter-hall cultural competition at IIT Kanpur* March 2023
 - Represented Hall 3 and won 1st position in Book Club's Scavenger Hunt and Pitch Meeting competitions